Literal structural certificates for $L_3(5)$ and $L_3(5) : 2$

L054 recovery component

Theorem 1. *For each of the 348 rows in the two fixed current companion tables below, the actual subgroup representative in the acquired numerical catalogue admits the displayed recursive structural description, with the source-qualified meaning of the operations used in the accepted naming components.*

Table	Group order	Rows	Native pages
$L_3(5)$	372000	140	6
$L_3(5) : 2$	744000	208	8

The tables are attributed to Thomas Connor and Dimitri Leemans. These are the currently retained companion PDF editions, not an assertion of identity with an archived historical file. All fourteen native pages were directly read. The accepted numerical row bijections identify the exact embedded groups; no group is identified by its order alone.

A colon asserts an actual normal subgroup and a complement, including a nontrivial action; a dot asserts an actual nonsplit extension. A direct-product symbol additionally requires commuting factors. Unparenthesized operations associate to the left. Explicit right parentheses are retained. The redundant parentheses around 4^2 in the indicated $5^{1+2} : (4^2)$ rows and around D_{10} in $\mathrm{Q}_8 : (D_{10})$ do not change their atomic meanings. In particular, source row 106 of the first table and row 127 of the second print the cyclic group 24, not 2^4. Powers and the extraspecial notation have their acquired literal interpretations; no unstated exponent or extraspecial sign is inferred. These labels need not determine a unique isomorphism type.

Acquired premises and actual models. The complete numerical catalogues and their source bijections, the earlier literal registry models, the generator-edge isomorphism criterion, and the previous split, direct, prime-index and small-quotient extension verifiers are acquired through the exact manifests in `dependencies.json`. The newly named atom $L_3(5)$ is the actual 31-point group of the retained clean `PSL(3,5)` execution. Its original ordered generators equal those of the acquired numerical top group. The classical constructor's named-group identification is retained explicitly, not inferred from the number 372000. The accepted 5^{1+2} and $2 \cdot A_5$ models and their elementary proofs are reused. The central commuting-lifts criterion includes the root's previously disclosed contribution; it is not a new argument of this component.

Every transport fixes the identity, is a bijection onto the actual group, and preserves right multiplication by every ordered model generator. Induction on words proves it is a homomorphism. The complete word-map format stores generator images and regenerates the entire bijection, testing every Cayley edge; no sample of words substitutes for this condition.

A smaller faithful action with the original embedding retained. The $L_3(5) : 2$ catalogue acts on 744 points. Its actual source-row-3 subgroup H has order 12000, hence index 62. The acquired right-coset method supplies 62 representatives t_i, starting at the identity, with explicit words in the original ordered generators. Literal enumeration of H verifies its order and $t_i t_j^{-1} \notin H$ for distinct representatives. For every ambient and every saved subgroup generator g, the proposed image p_g is a permutation and satisfies

$$t_i g t_{p_g(i)}^{-1} \in H \quad (1 \le i \le 62).$$

Thus it is the actual action on right cosets Ht_i. Literal closure of the image has the acquired ambient order 744000, so the action is faithful. All subgroup-image arrays are checked using the same equation. This proves the original embedded representatives are transported to the groups used by the structural reader; it is not a replacement by a convenient abstract model. The first table keeps its original 31-point action.

Lemma 1 (Complete quotient-generator lifts)**.** *Let $N \lhd X$ be finite. Let Q be a finite group with ordered generators $q_1, \dots, q_k$, and let $\alpha : Q \longrightarrow X/N$ be a verified isomorphism. Choose $x_j \in X$ lifting $\alpha(q_j)$. Then N has a complement in X if and only if at least one of the $|N|^k$ ordered choices*

$$y_j \in N x_j \quad (1 \le j \le k)$$

extends to a homomorphism $Q \to X$. Such an extension is decided by testing all edges of the complete ordered Cayley graph of Q.

Proof. A complement gives a section of the quotient map and hence one of these choices. Conversely, if $s : Q \to X$ extends a choice, its composition with the quotient map agrees with α on the generators, and therefore on all of Q. Consequently s is injective, its image meets N trivially, and the image has index $|N|$ in X; it is a complement.

For the decision procedure, list every element of Q by breadth-first multiplication in the ordered generators, with the identity first. Assign identity image, and propagate the proposed images along every generator edge. A conflicting image proves that the assignment is not a homomorphism. If there is no conflict, every group element has an image and every generator edge is preserved. The same induction on words proves a homomorphism. The list is complete, so all relations needed for the conclusion have been checked, without choosing or assuming a presentation. Every generator lift has the form nx_j, so the finite product of these lists omits no section. □

Application and literal verification. For the actual $Q_8 \cdot A_5$ and $Q_8 \cdot (A_5 : 2)$ models, the reader first verifies the normal quaternion subgroup by its acquired model transport. It partitions the actual ambient elements into disjoint complete normal cosets and computes the right quotient action on those cosets. An explicit generator-edge bijection identifies that action with an already verified model of the printed quotient. Each proposed lift has precisely the stated quotient image. The first quotient model has three ordered generators, the second two; all $8^3 = 512$ and $8^2 = 64$ choices respectively fail a complete-edge homomorphism test. The lemma proves both displayed nonsplittings. No complement-nonexistence result returned by the witness supplier is a premise of the literal checker.

All other new registry entries have the acquired elementary atom, normal-subgroup, complement or earlier nonsplitting certificates. References are checked in dependency order. For each of the 348 source rows the reader checks the exact source text, native abstract syntax, numerical row map, original generators, actual subgroup order and complete transport. The source and numerical counts are not filters in a new subgroup search.

Representation and finite controls. The reader uses byte arrays for permutation degrees from 31 through 256, and tuples otherwise. For zero-based arrays the right-action product is $(ab)[i] = b[a[i]]$. Byte translation by the array of b, padded only beyond its degree, computes exactly this formula, since every entry of a is strictly below that degree. Encoding therefore changes neither the product nor the complete closure and edge tests. Independent tuple-oracle controls check every pair of permutations of four points embedded at degrees around both representation boundaries, including the actual degrees 31 and 62. Small exact coset controls include a nonnormal subgroup, wrong orientation and a nonfaithful quotient action. The new lift criterion is checked against a separately enumerated complete subgroup lattice for five small split and nonsplit extensions, including a nonabelian quotient. Damaged actual quotient/lift and source/action inputs must be rejected. Ordinary and optimized Python executions must give the same full report.

Scope and authorship. The source descriptions belong to Connor and Leemans. The present component adds literal applications, the complete quotient-generator lift check and its controls, and the actual 62-coset transport. Prior accepted methods, models, classifications and reviews retain their original attribution. GAP is used only to propose witnesses in saved actual groups. The final replay is independent literal permutation arithmetic and does not rerun that witness search or an ambient subgroup census. This is a component for 348 current-table descriptions, not a complete L054 recovery, a canonical naming theorem, a historical byte-identity claim, or a new measurement.